

8T1: Spectral-based sound transformations (1 of 2)

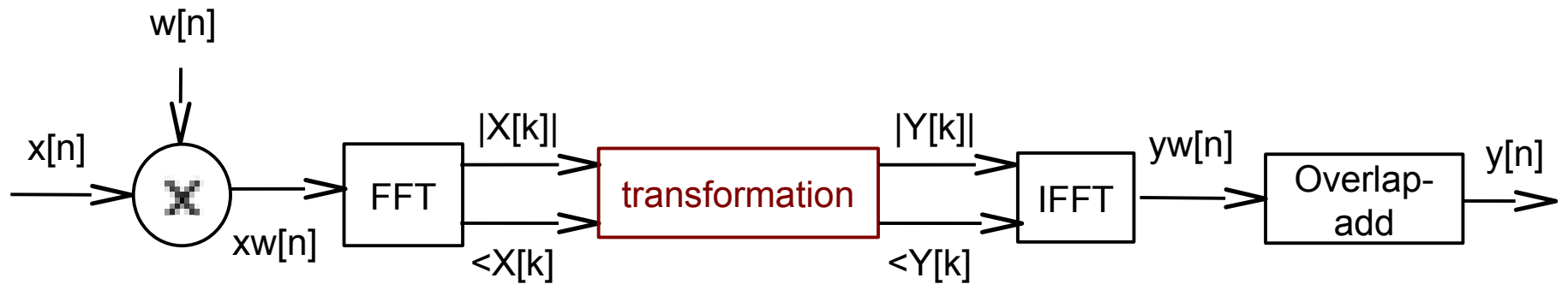
Xavier Serra

Universitat Pompeu Fabra, Barcelona

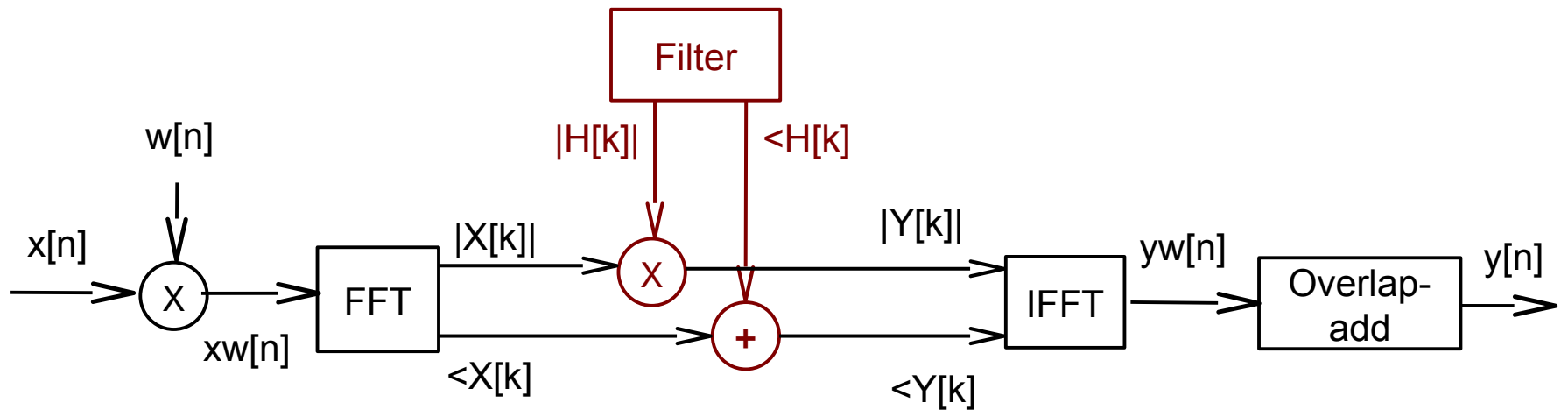
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Short-time Fourier transform

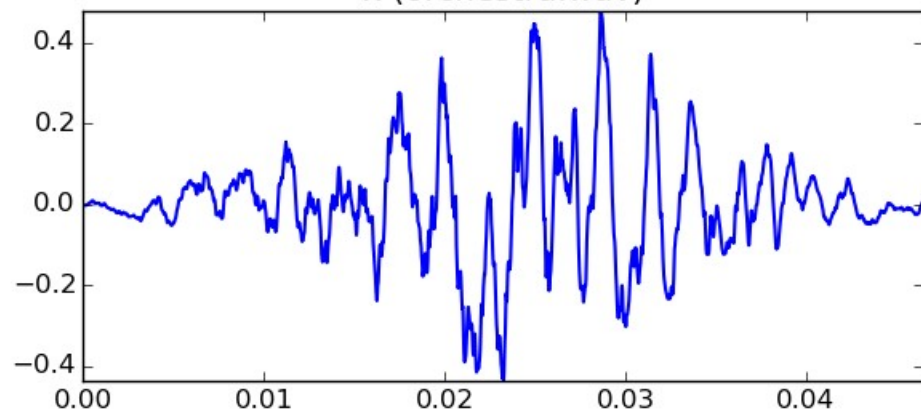


Filtering with STFT

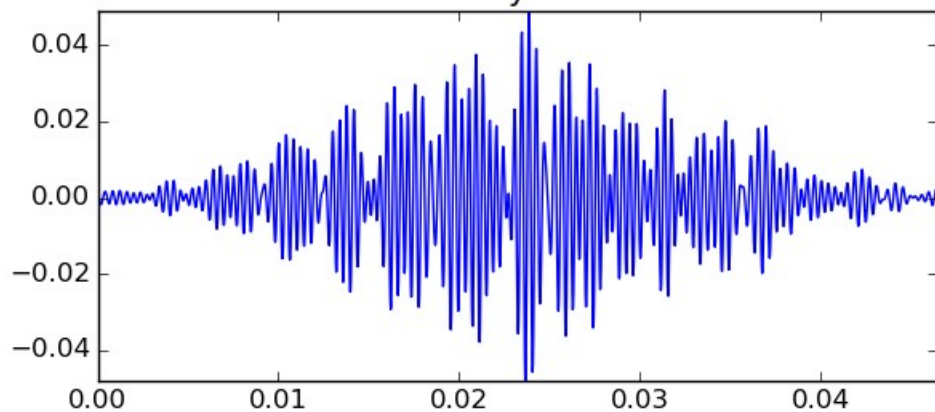


$$Y_l[k] = |H[k]| |X_l[k]| e^{j(\angle H[k] + \angle X_l[k])}$$

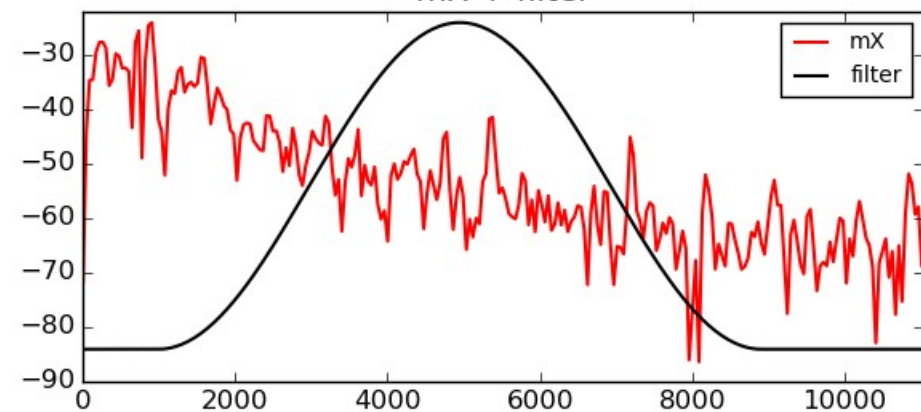
x (orchestra.wav)



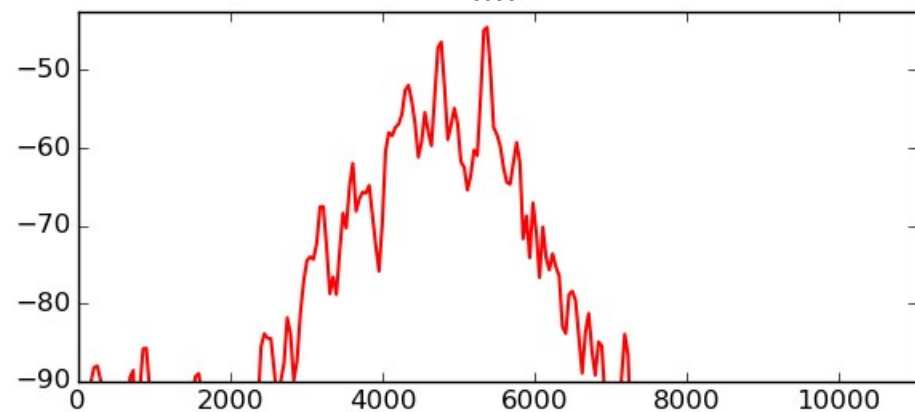
y



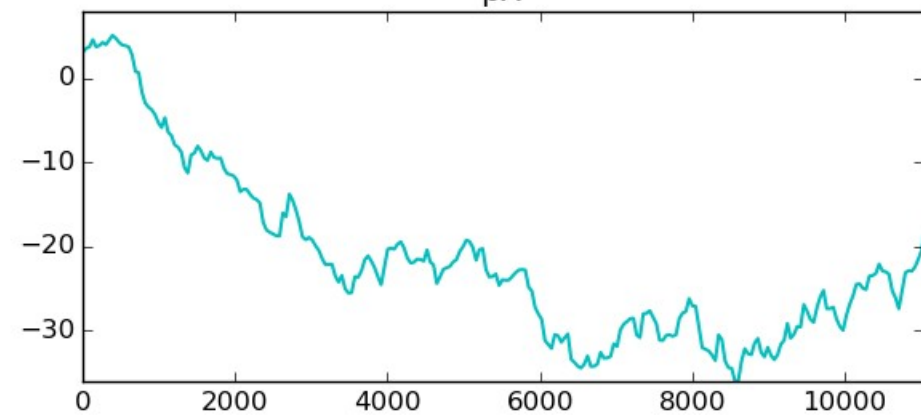
mX + filter



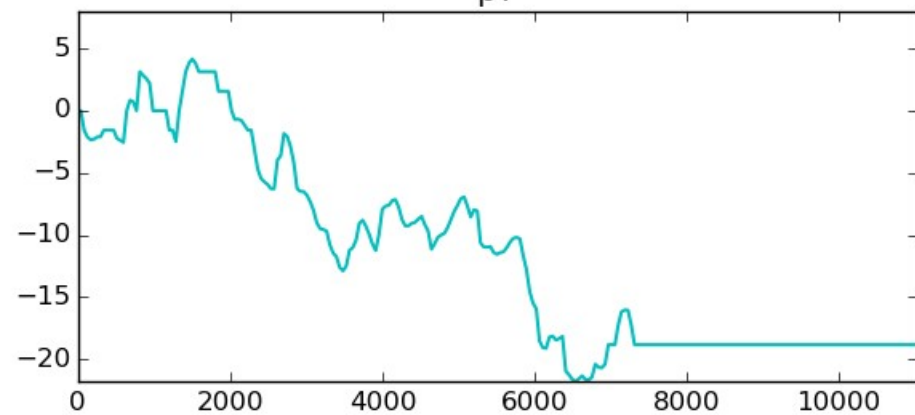
mY



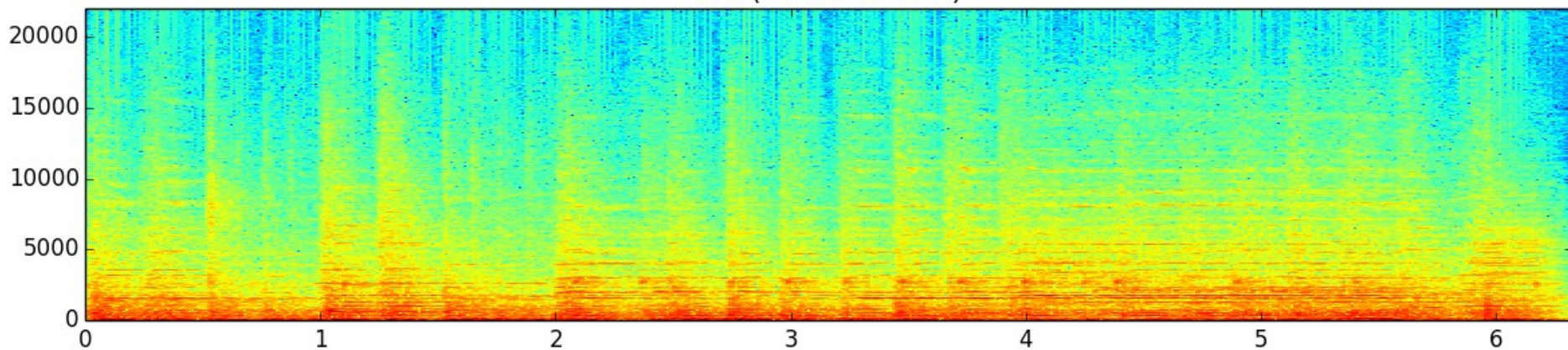
pX



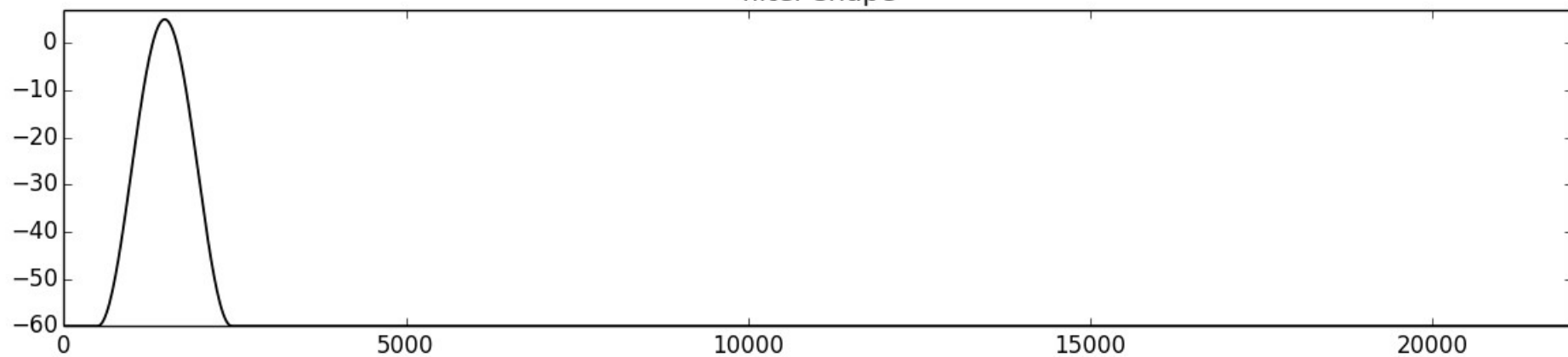
pY



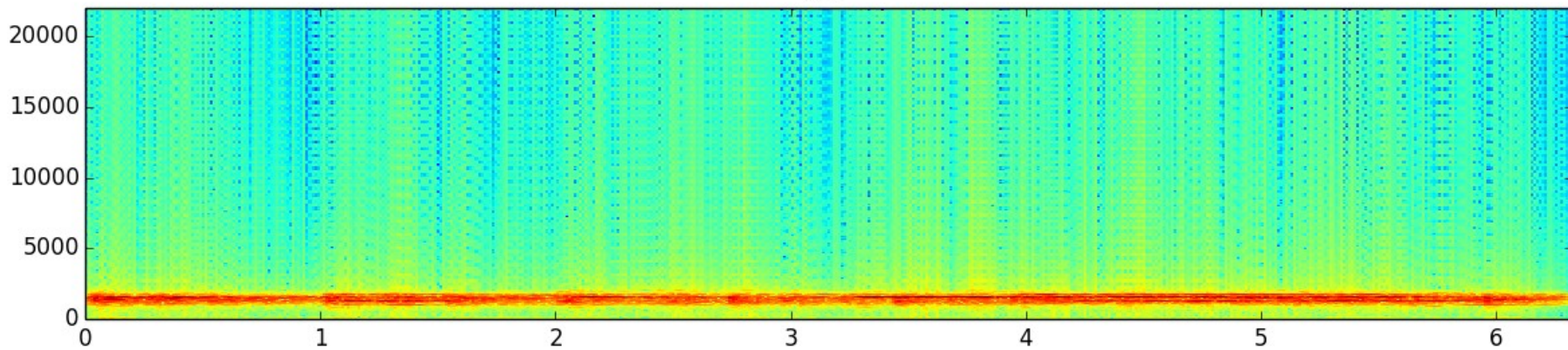
mX (orchestra.wav)



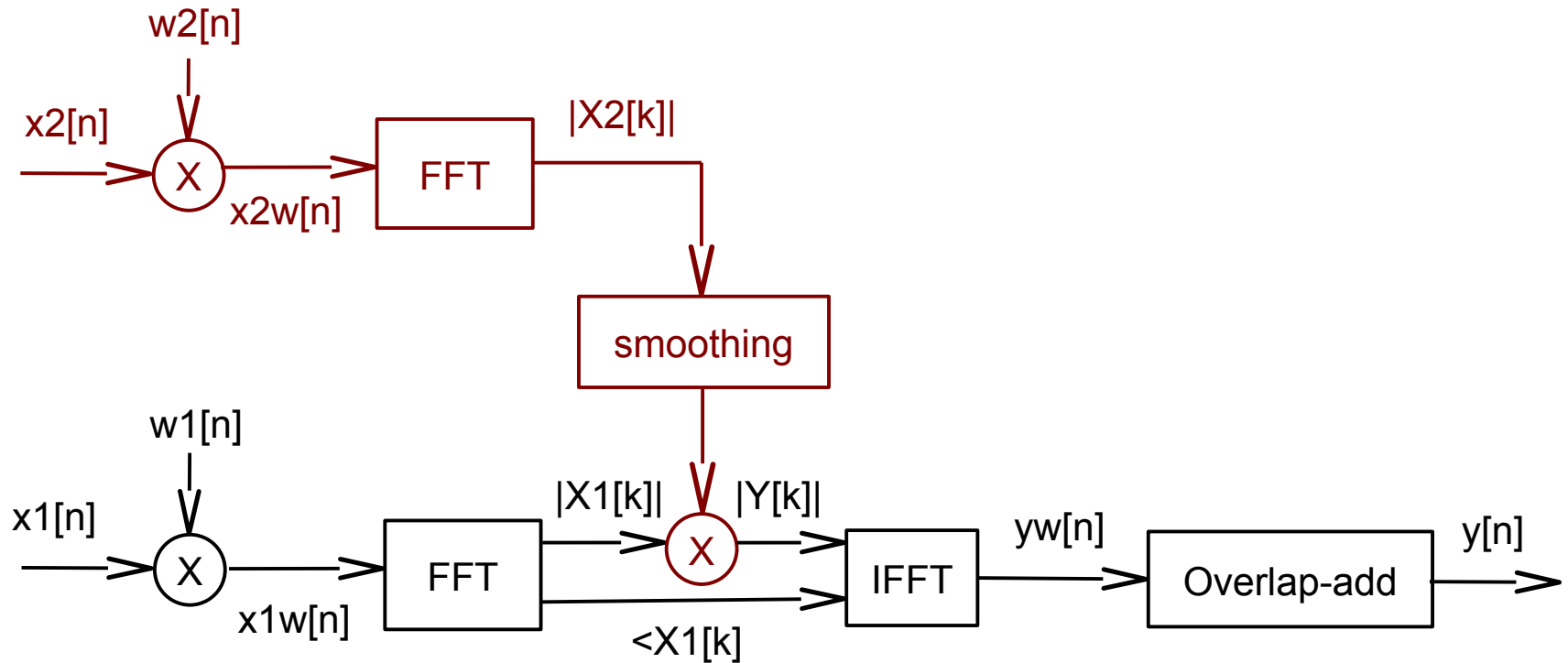
filter shape



mY

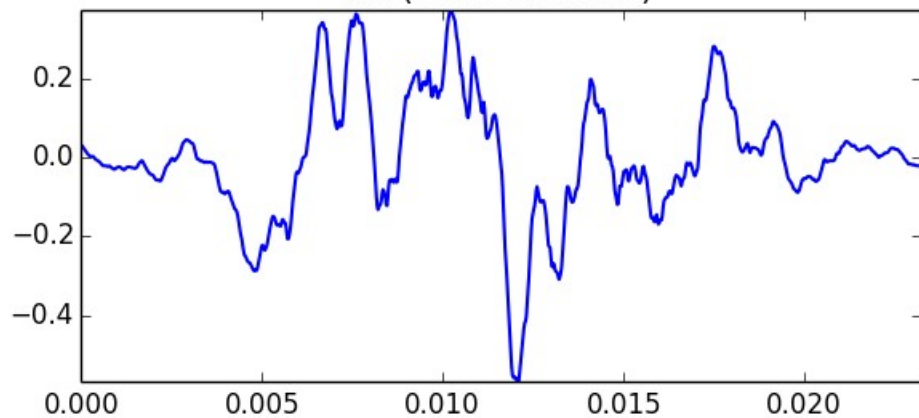


Morphing with STFT

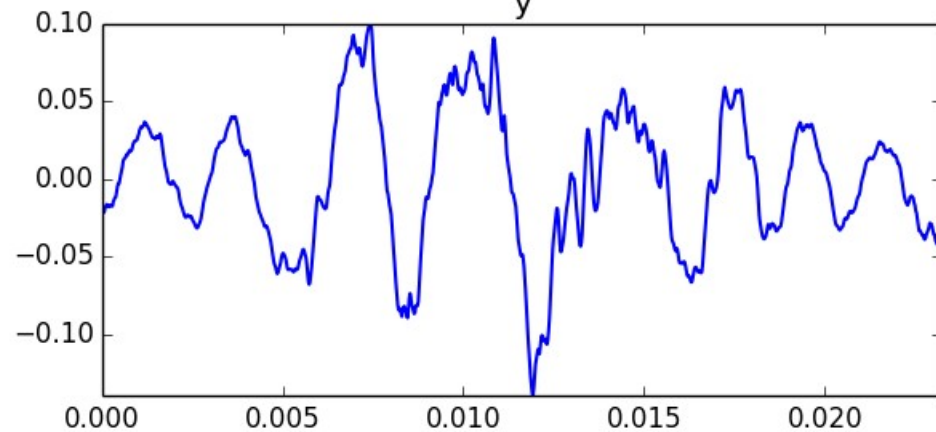


$$Y_l[k] = |X_{2_l}[k]| |X_{1_l}[k]| e^{j \angle X_{1_l}[k]}$$

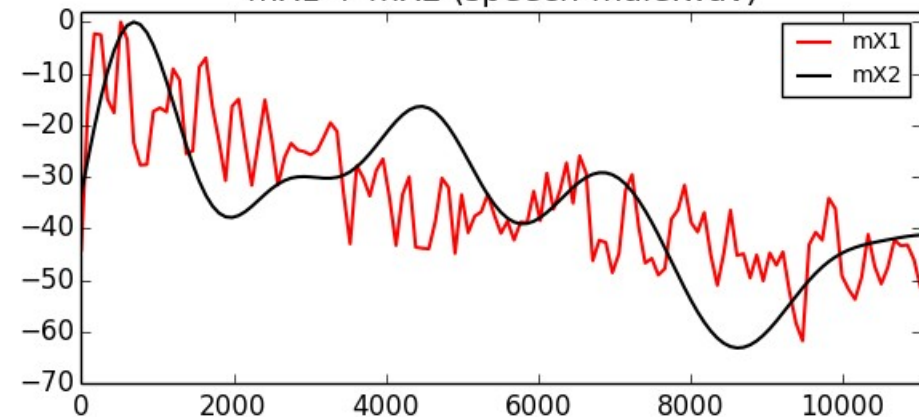
x1 (orchestra.wav)



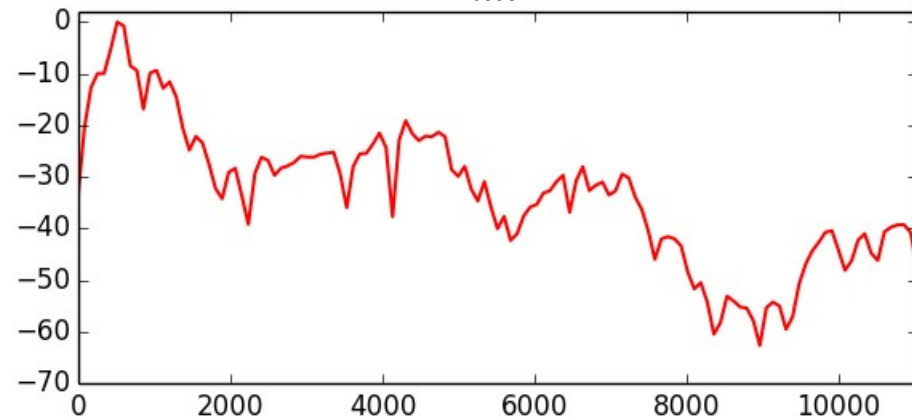
y



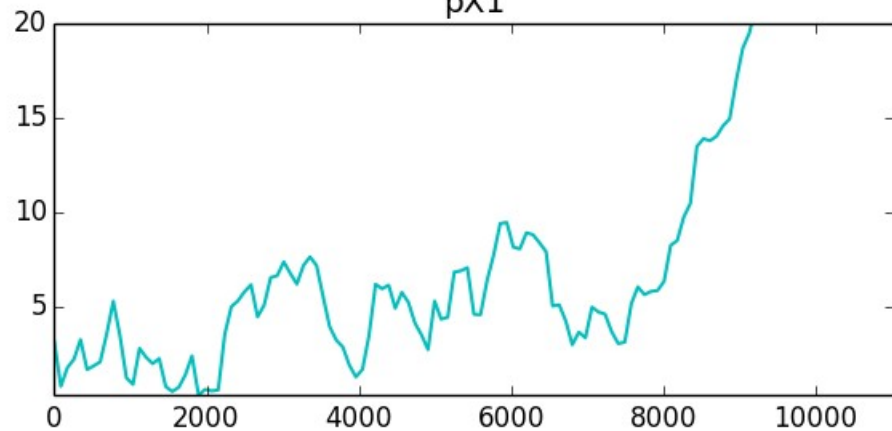
mX1 + mX2 (speech-male.wav)



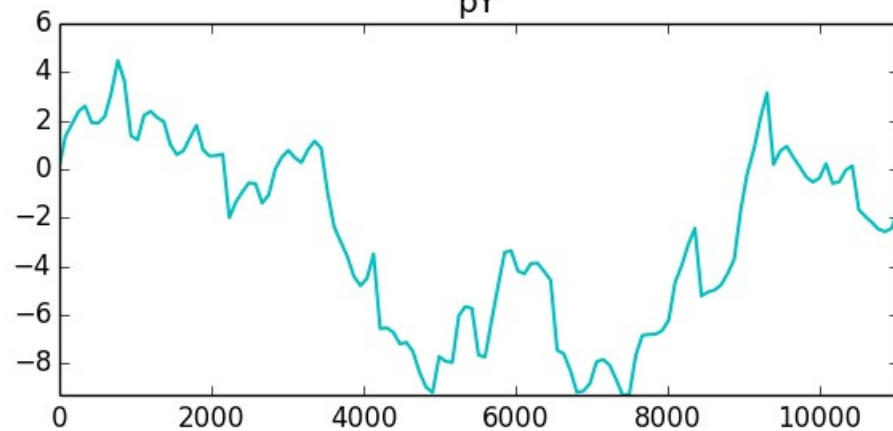
mY

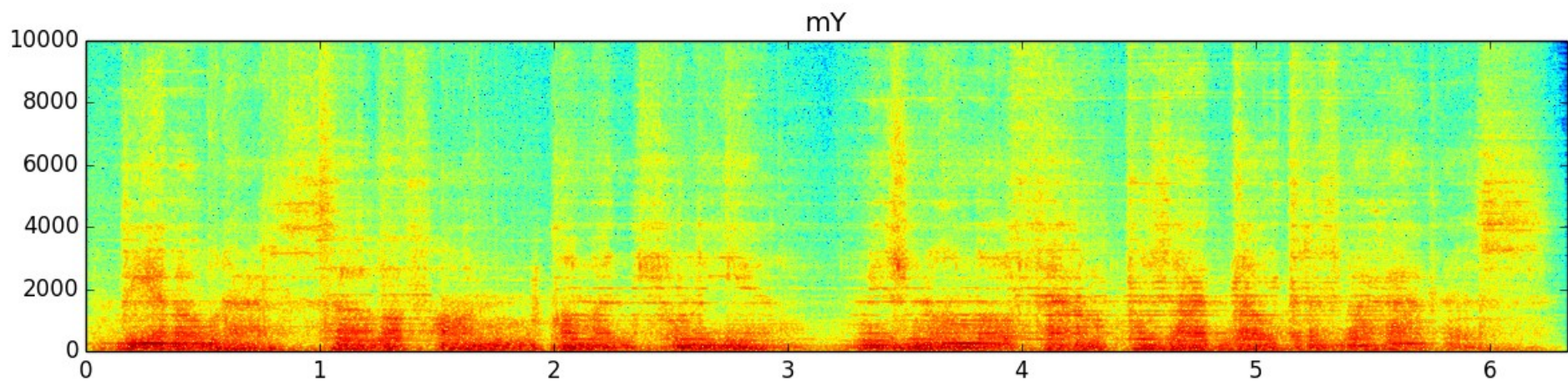
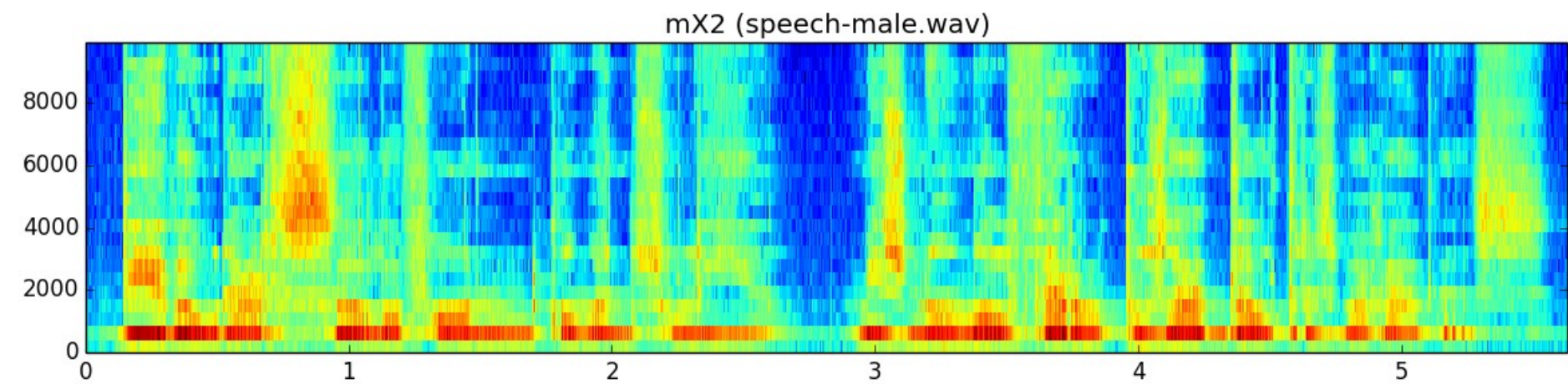
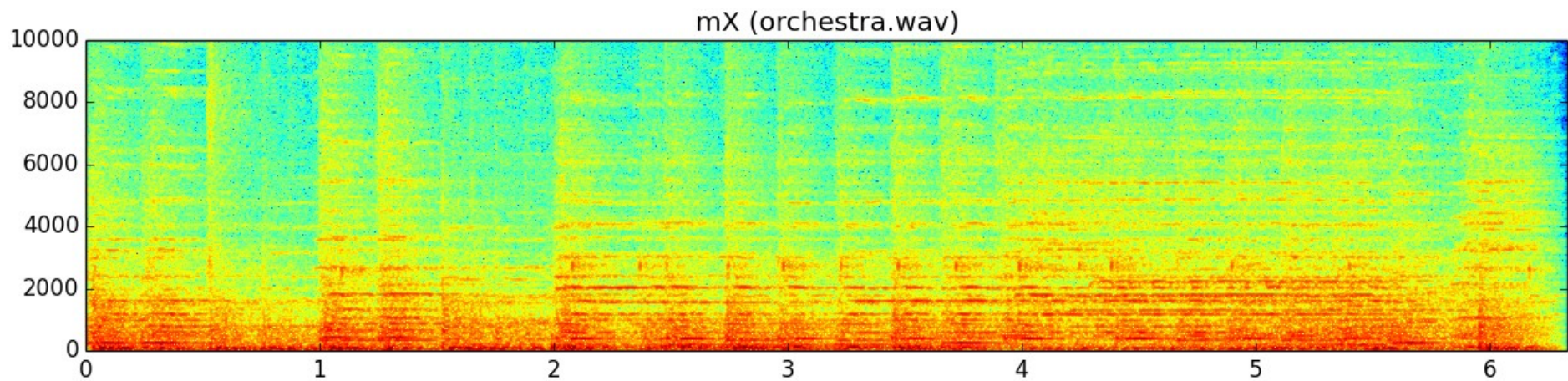


pX1

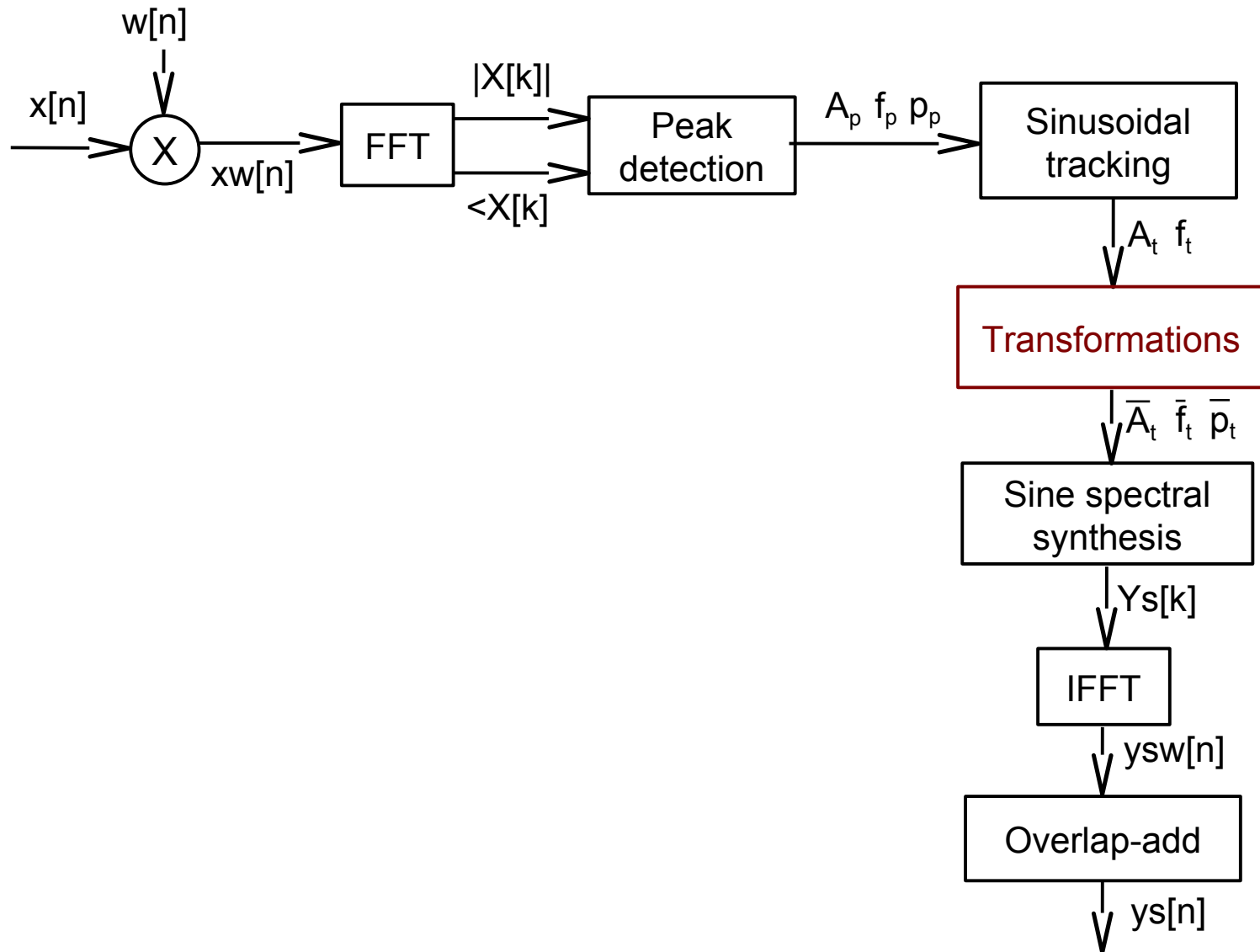


pY





Sinusoidal model



Scaling frequency, amplitude and time

$$\bar{f}_t[q] = sf_t[l] f_t[st_t[l]l]$$

$$\bar{A}_t[q] = sA_t[l] + A_t[st_t[l]l]$$

$$\bar{\phi}_t[q] = \phi_t[q-1] + f_t[q]$$

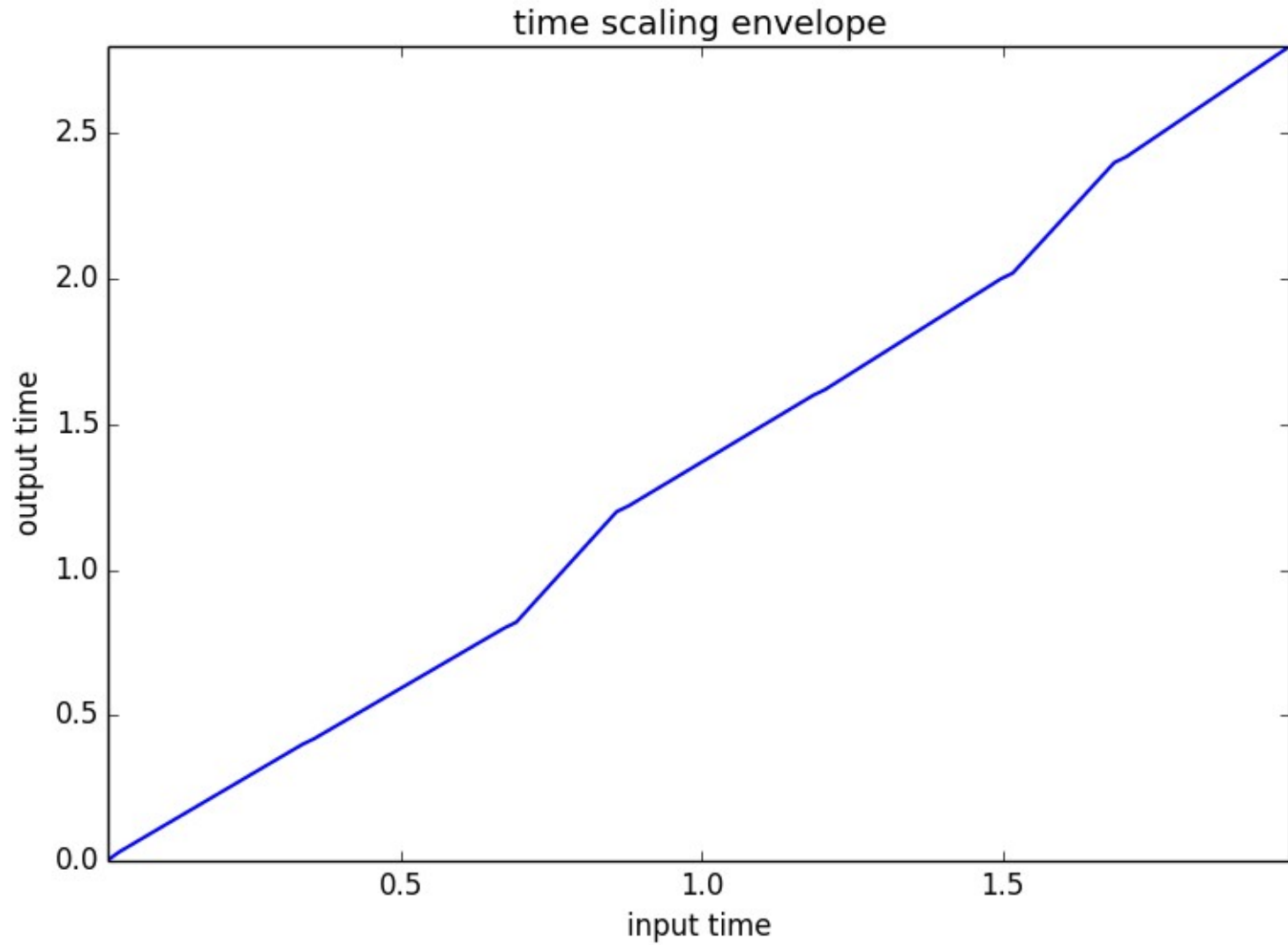
q : output frame index ; l : input frame index ; t : sinusoidal track index

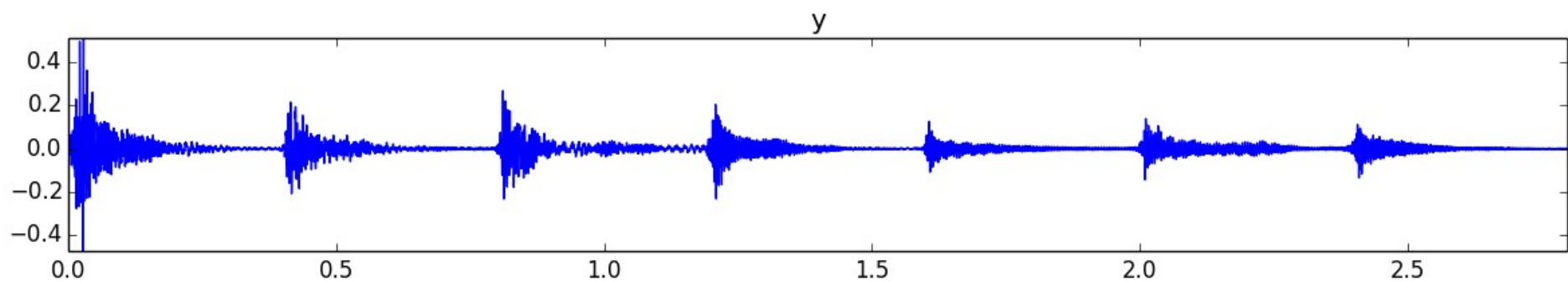
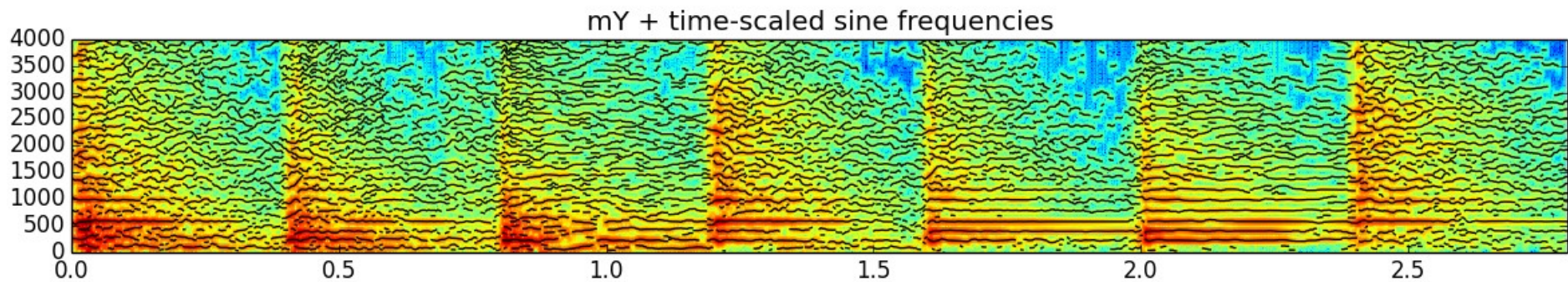
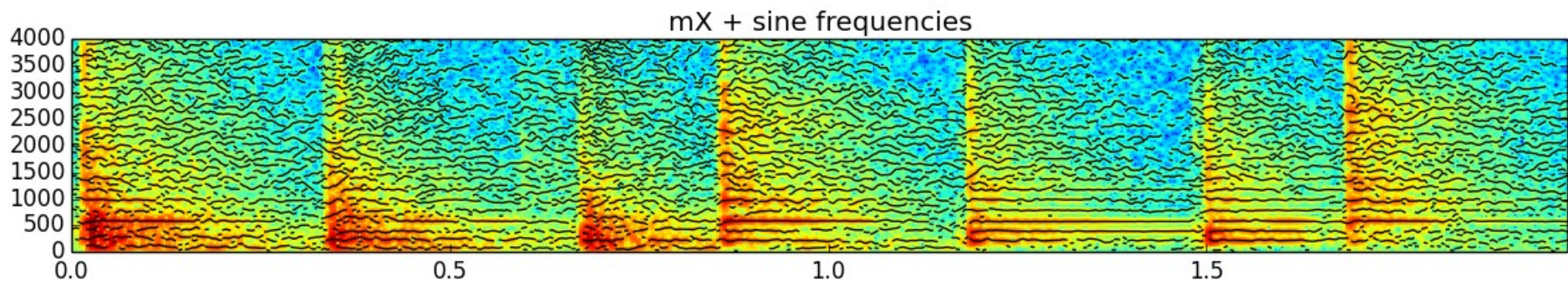
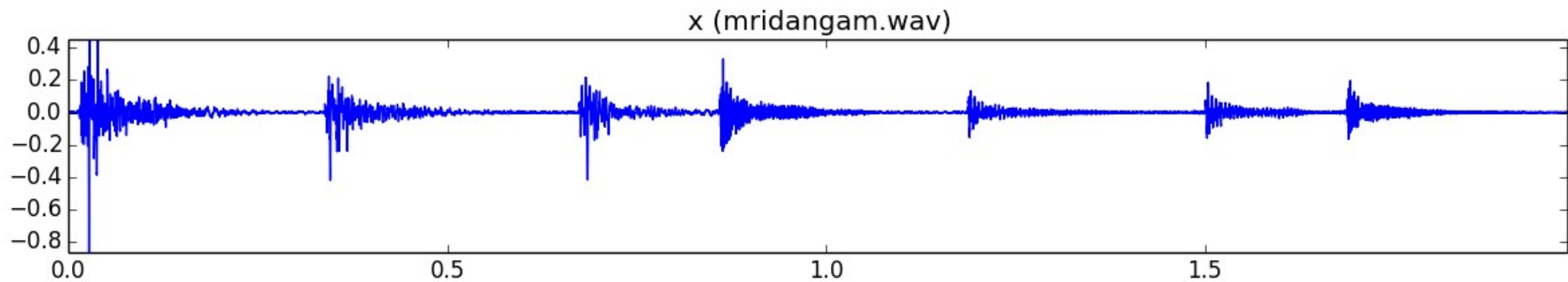
f : input frequency in Hz ; A : input amplitude in dB

sf : scaling frequency ; sA : scaling amplitude ; st : scaling time

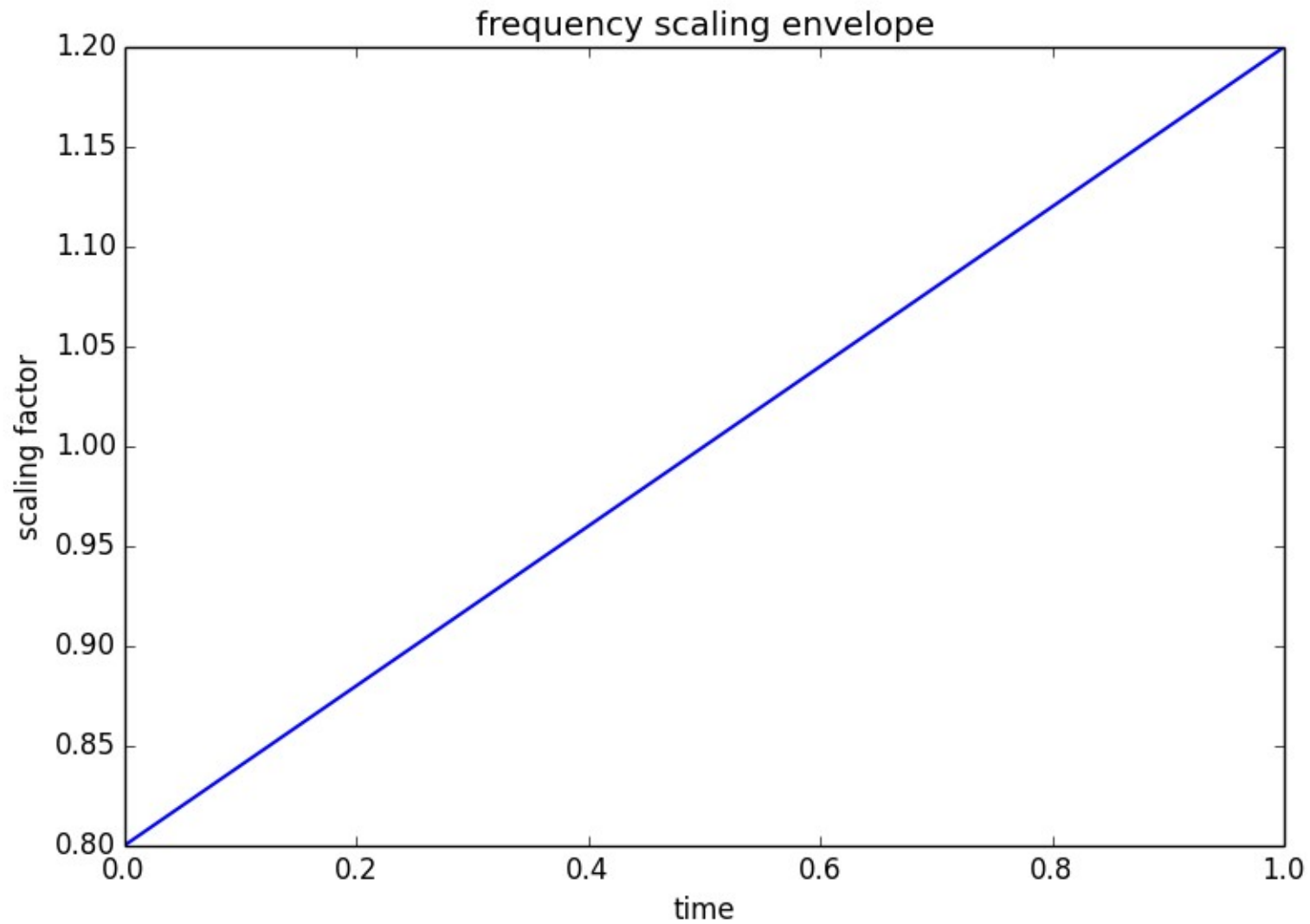
$\bar{f}$: output frequency ; $\bar{A}$: output amplitude ; $\bar{\phi}$: output phase

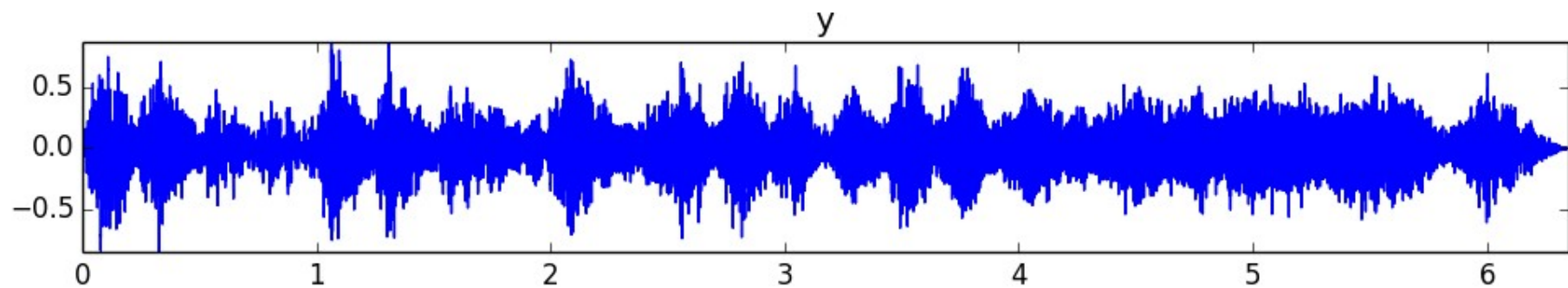
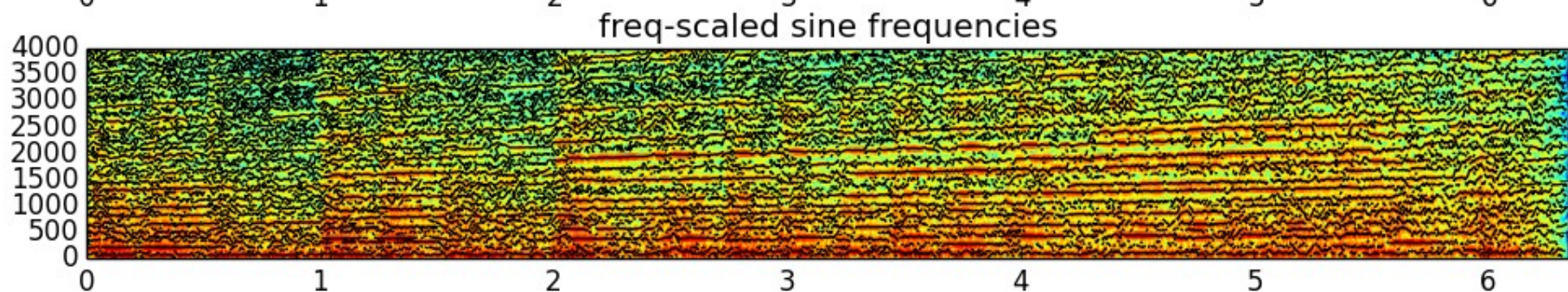
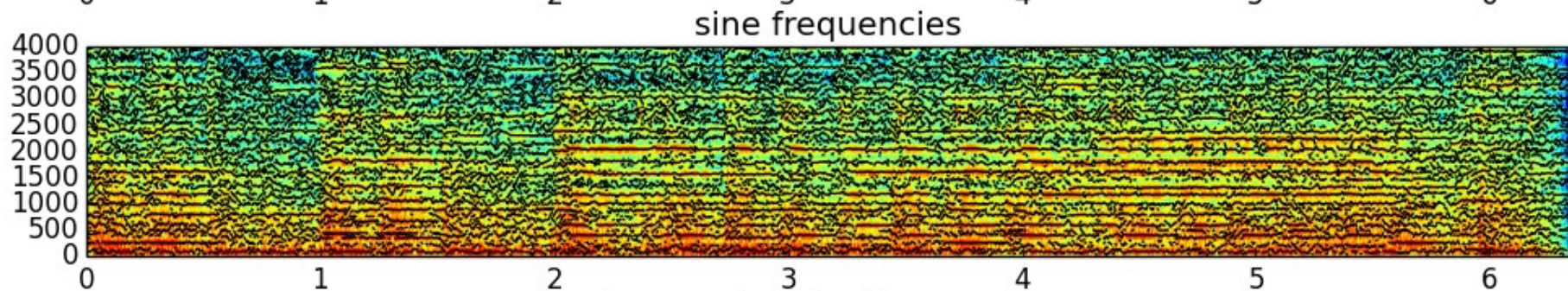
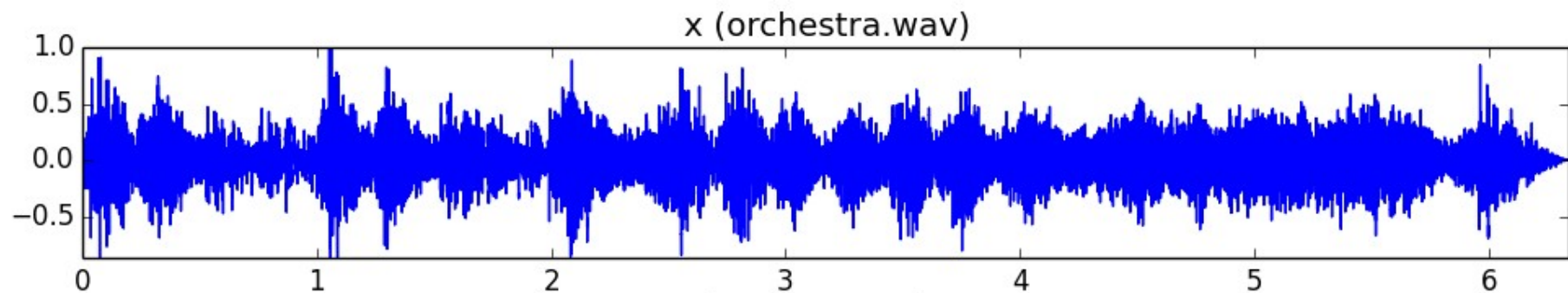
Time scaling





Frequency scaling





References

- More information on this topic from Wikipedia:
 - http://en.wikipedia.org/wiki/Sound_effects
 - http://en.wikipedia.org/wiki/Equalization_filter
 - http://en.wikipedia.org/wiki/Audio_timescale-pitch_modification
- Sounds:
<http://www.freesound.org/people/xserra/packs/13038/>
- Slides released under CC Attribution-Noncommercial-Share Alike license and code under Affero GPL license; available from <https://github.com/MTG/sms-tools>

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&

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